Yihang **Zuo**

(+1) 602-758-9199 | yihangzu@asu.edu | zyh0911.github.io | github.com/zyh0911 | linkedin.com/in/yihang-zuo-69222b259

Education

Arizona State University Ph.D. in Electrical and Computer Engineering — Tempe, AZ. Aug 2025 ∼ Present*.

ECE Ph.D. student advised by Prof. Deliang Fan, working on efficient AI systems across the stack: KV-cache and inference efficiency for multimodal agents, on-device LLM adaptation, and memory-centric accelerators. Papers at ICCAD, ISQED (Best Paper Award), NeurIPS, DAC, DATE and ISLPED. *Seeking a Summer 2027 internship; expected graduation May 2029.

HKUST (Guangzhou) M.Phil. in Microelectronics — Guangzhou, China. 2023 ∼ 2025.

South China University of Technology B.E. in Microelectronics — Guangzhou, China. 2019 ∼ 2023.

Experience

Doctoral Researcher @ Arizona State University

Tempe, AZ. Aug 2025 ∼ Present.

1. **3D memory-centric accelerator design** — Papers #3, #4
   - **Device-architecture co-design**: modified the HSPICE compact model to fit measured BEOL MoS_2 transistors and built a 2T0C monolithic 3D gain-cell eDRAM on top of it, with device-to-architecture modeling and evaluation (Under review).
   - **Algorithm-architecture co-design**: exploited the fast, low-power eDRAM readout to design a monolithic 3D lookup-table accelerator with dual-side compression, taken from Verilog RTL to logic synthesis in TSMC 28 nm (Under review).
   - **Architecture-scheduling co-design**: designing a 3D DRAM accelerator for vision-language-action (VLA) workloads, jointly optimizing the dataflow scheduling and the near-memory circuit design (In progress).
2. **Runtime KV-cache management for computer-use agents** — In progress (**DAC 2027**)
   - Designing a cross-step cache: incremental ViT re-encodes only changed windows (26–50% ViT FLOPs saved with no significant accuracy loss on AgentNetBench), plus historical screenshot KV reuse with position remapping and selective recomputation to cut TTFT.
   - Proposed GQA-aware KV compression with a two-stage frame/token budget: at a 20% budget it reaches 21.1% step accuracy vs. 16.5% for the best baseline (GUI-KV) with 5.7× less KV memory.
3. **Efficient on-device learning for LLMs and edge accelerators** — Papers #1, #5
   - Designed programming-free multi-task adaptation on ReRAM CIM: a crossbar column-wise step mask applied to partial sums through the existing shifter-adder path keeps the weights fixed, using 0.03% of fine-tuning energy with 3.9% area overhead and +3.6% accuracy on DeiT (**ICCAD**).
   - Showed that zeroth-order LLM fine-tuning is dominated by a single decoding layer that can be located before training; tuning it alone matches full-model ZO on LLaMA2-7B and Qwen3-8B with up to 4.52× speedup (**NeurIPS**).
   - Developed memory-efficient sparse forward-only continual learning for on-device adaptation without backpropagation (Under review).
4. **LLM agents for chip physical design** — In progress (**DAC 2027**)
   - Extending CoEvoP&R, an LLM framework that evolves readable, differentiable placement objectives in DREAMPlace with an archive-guided (MAP-Elites) search, to the full OpenROAD flow (placement, CTS, routing), so objectives and stage parameters are co-evolved against post-route PPA instead of placement-stage surrogates.
   - Built a bottleneck-aware agentic loop: stage reports (WNS/TNS, congestion, utilization, DRC) are compressed into a compact state, the LLM proposes edits aimed at the dominant QoR bottleneck, and a certification step accepts an edit only if it improves stage QoR without regressing downstream metrics.

Graduate Researcher @ HKUST (Guangzhou)

Guangzhou, China. 2023 ∼ 2025.

1. **Hardware-mapping co-design for compute-in-memory ViT accelerators** — Paper #2 (**ISQED**)
   - Built Harmony to co-optimize CIM macro configuration (crossbar size, macro count, ADC precision) and the spatial mapping of attention/MLP layers: 48% area, 13% latency and 32% energy reduction (**ISQED Best Paper**).
   - Co-designed a heterogeneous RRAM/SRAM CIM accelerator with a hybrid dataflow and in-network partial-sum reduction: 3.57× speedup and 2.78× energy efficiency over homogeneous digital CIM (**ISLPED**).

Also contributed to an open-source digital CIM compiler (DATE 2025) and LLM-powered multi-agent writing systems (CHI 2026, DIS 2026).

Research Intern @ Shanghai AI Laboratory

Shanghai, China. Sep 2022 ∼ Mar 2023.

Co-developed OpenDRC, a GPU-accelerated design rule checking engine: 4.5× faster than the prior GPU checker X-Check and 9.6–13× faster than KLayout (**DAC 2023**).

Selected Lead-Author Publications

1. **PRISM: a programming-free on-device multi-task adaptation framework for ReRAM CIM** **ICCAD** 2026.
   Yihang Zuo, Wanhao Yu, Asmer Ali, Li Yang, Deliang Fan
   Adapts a deployed ReRAM accelerator to new tasks with column-wise masks instead of rewriting weights.
2. **Harmony: hardware-mapping co-exploration for hybrid CIM-based ViT accelerators** **ISQED** 2026 (Best Paper).
   Yihang Zuo, Zexin Fu, Cong Wang, Yuchao Wu, Jiayi Huang, Yuzhe Ma
   Jointly searches CIM macro configuration and layer mapping, cutting area by 48% and energy by 32%.
3. **MOSS: monolithic 3D embedded memory based on 2T0C MoS_2 gain-cell eDRAM** Under review.
   Yihang Zuo, et al.
   Device-to-architecture modeling of BEOL MoS_2 gain-cell eDRAM for monolithic 3D integration.
4. **LUMOS: a monolithic 3D MoS_2 eDRAM-based lookup-table accelerator with dual-side compression** Under review.
   Yihang Zuo, et al.
   Uses fast, low-power eDRAM readout to build a lookup-table inference accelerator in TSMC 28 nm.
5. **FlyCL: memory-efficient sparse forward-only continual learning** Under review.
   Yihang Zuo, et al.
   Continual learning on edge devices without backpropagation and with a small memory footprint.

Other Relevant Publications

Asterisks (∗) denote lead-author papers.

Dominant-Layer ZO: a single layer dominates zeroth-order fine-tuning of LLMs **NeurIPS** 2026.
Vistoria: multimodal fictional story writing through image-text co-editing **CHI** 2026.
Spatial Balancing: an LLM-powered spatial interface for science communication writing **DIS** 2026.
Optimizing heterogeneous CIM with hybrid dataflow and in-network reduction for ViT **ISLPED** 2025.
OpenC^2: an open-source end-to-end hardware compiler framework for digital CIM macros **DATE** 2025.
OpenDRC: an efficient open-source DRC engine with hierarchical GPU acceleration **DAC** 2023.
∗ Symmetrical indoor visible light layout optimized by a modified grey wolf algorithm Applied Optics 2022.

Awards and Services

- **Best Paper Award** Harmony at ISQED 2026. 2026.
- **Reviewer** AAAI 2027; IEEE JETCAS 2026; ASP-DAC, DATE and DAC 2025. 2025 ∼ 2027.
- **Postgraduate Studentship** HKUST (Guangzhou). 2023 ∼ 2025.
- **Meritorious Winner** Mathematical Contest in Modeling (MCM). 2021.
- **1st Prize (South China)** China College IC Competition. 2022.

Skills

- **Programming** Python, C/C++, CUDA, Verilog, Shell, MATLAB; Linux, Git, Docker, Slurm.
- **Efficient LLM** PyTorch, HuggingFace Transformers; quantization, pruning, LoRA/PEFT, zeroth-order and forward-only fine-tuning; KV-cache reuse and compression, prefill/TTFT and FLOPs profiling of multimodal agents.
- **GPU Computing** CUDA kernels for computational geometry (edge-based DRC checks), hierarchical GPU parallelization, performance analysis against CPU baselines.
- **Architecture** Accelerator and memory modeling (NeuroSim, CACTI, NVSim, DESTINY); dataflow, NoC and mapping design-space exploration; analytical PPA models; 3D DRAM/eDRAM and compute-in-memory.
- **Hardware & EDA** OpenROAD, DREAMPlace, Yosys/ABC, KLayout; Synopsys VCS, Design Compiler, PrimeTime (STA); Cadence Innovus, Virtuoso; Siemens Calibre; HSPICE; Vivado/HLS (FPGA); ASAP7, TSMC 28 nm.
- **Languages** Chinese (native), English (fluent).